

FISHERGEOMETRICS FRAMEWORK

Derivation of $\tau = \frac{1}{5}$ and $\kappa = \frac{6}{5}$ from the Einstein Metric on $M^{1,1,1}$

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Main result. The FisherGeometrics parameters $\tau = \frac{1}{5}$ and $\kappa = \frac{6}{5}$, previously postulated, are uniquely determined by the Einstein metric on the coset space $M^{1,1,1} = \frac{SU(3) \times SU(2) \times U(1)}{SU(2) \times U(1) \times U(1)}$. They follow from the unique real positive root $\beta^* = \frac{1}{4}$ of the cubic equation (4.15) of Castellani, D'Auria and Fré (1984), together with eq. (4.16) giving $\alpha^* = \frac{1}{2}$.

1. Background

The FisherGeometrics framework (Bouwman 2026) is built on the postulate $g_{AB} = \mathcal{F}_{AB}/\rho_0$ and uses two parameters beyond the single measurement G_N :

$$\tau = \frac{1}{5} \quad (\text{BGK relaxation / radius of } S^1 \subset K) \tag{1}$$

$$\kappa = \frac{6}{5} \quad (\text{holonomy coupling / radius of } \mathbb{CP}^2 \subset K) \tag{2}$$

These parameters appear throughout the framework:

Observable	Formula	Role of τ, κ
G_F	$\tau^2(n^2 + 1)/(4M_{\text{KK}}^2)$	τ^2
η_{baryon}	$(3/4)\tau^{13}$	τ^{13}
Λ_{QCD}	$M_{\text{KK}}e^{-2\pi}(1 - \tau)^2$	$(1 - \tau)^2$
λ_W	$\tau\sqrt{\kappa}$	$\tau\sqrt{\kappa}$
$\text{vol}(S^1)$	$2\pi\tau$	τ
$\text{vol}(S^3)$	$2\pi^2\tau^3$	τ^3

The question is whether τ and κ are independent postulates or consequences of the geometry of the internal space $K = \mathbb{C}P^2 \times S^3 \times S^1$.

2. The internal space $M^{1,1,1}$

The isometry group of K is $G_K = SU(3) \times SU(2) \times U(1)$, which is the Standard Model gauge group. The internal space is the coset space

$$M^{1,1,1} = \frac{SU(3) \times SU(2) \times U(1)}{SU(2) \times U(1) \times U(1)} \quad (3)$$

with $\dim(M^{1,1,1}) = 12 - 5 = 7$.

This space was classified by Castellani, Romans and Warner (1984) as the unique member of the family $M^{p,q,r}$ that admits $\mathcal{N} = 2$ supersymmetry (holonomy group $SU(3)$, 2 Killing spinors). All other $M^{p,q,r}$ with $p \neq q$ have holonomy $SO(7)$ and no supersymmetry.

Theorem 1 ($\mathcal{N} = 2$ uniqueness). *Among all seven-dimensional coset spaces with $SU(3) \times SU(2) \times U(1)$ isometry, the unique space admitting two Killing spinors ($\mathcal{N} = 2$ supersymmetry in 4D) is $M^{1,1,1}$, corresponding to $p = q = r = 1$ in the embedding (3.1) of Castellani et al. (1984).*

3. The Einstein metric on $M^{1,1,1}$

3.1 Metric ansatz

The most general G_K -invariant metric on $M^{1,1,1}$ is (Castellani, D’Auria and Fré, Nucl. Phys. B239 (1984) 610):

$$ds_7^2 = a^2 ds_{\mathbb{C}P^2}^2 + b^2 ds_{S^2}^2 + c^2 \eta^2 \quad (4)$$

where $\eta = d\psi + p\omega_{\mathbb{C}P^2} + q\omega_{S^2}$ is the connection 1-form of the $U(1)$ fibration, and $a^2, b^2, c^2 > 0$ are the three squashing parameters.

3.2 Parametrisation

Following eq. (4.11) of Castellani et al., introduce new variables $\alpha, \beta, \gamma > 0$:

$$a = \sqrt{\frac{q}{p}} \alpha, \quad b = \gamma \beta, \quad c = q\gamma \quad (5)$$

For $M^{1,1,1}$: $p = q = 1$, so $a = \alpha$, $b = \gamma \beta$, $c = \gamma$.

3.3 Einstein equations

The Ricci tensor of the metric (4) is computed via the Maurer–Cartan equations (4.1) and curvature formula (4.3) of Castellani et al. In the parametrisation (5), the Ricci components (eqs. 4.13a–4.13d) give the Einstein system $R_{\alpha\beta} = \nu g_{\alpha\beta}$ (where $\nu = 4e^2$ for the Freund–Rubin solution):

$$\beta(1 - \beta)\gamma^2 = \frac{3\nu}{4} \quad (6)$$

$$\alpha^2 - \frac{\beta^2}{p^2} q^2 \alpha + \dots = \frac{\nu}{4} \quad (7)$$

$$p^2 \alpha^2 \gamma^2 + \dots = \nu \quad (8)$$

3.4 The cubic equation

Eliminating α and γ , the system reduces to a single cubic in β (eq. 4.15 of Castellani et al.):

$$\boxed{4\beta^3 - 6\beta^2 + \left(\frac{9}{4} + \frac{q^2}{p^2}\right)\beta - \frac{1}{2}\frac{q^2}{p^2} = 0} \quad (9)$$

A key theorem (Castellani et al., p. 627) states that this cubic has *always exactly one real positive root* β^* , bounded by $0 < \beta^* < \frac{1}{2}$.

4. Explicit solution for $M^{1,1,1}$

4.1 The unique root β^*

For $M^{1,1,1}$: $p = q = 1$, so $q^2/p^2 = 1$. The cubic (9) becomes:

$$4\beta^3 - 6\beta^2 + \frac{13}{4}\beta - \frac{1}{2} = 0 \quad (10)$$

Solution: The unique real positive root is

$$\boxed{\beta^* = \frac{1}{4}} \quad (11)$$

Verification: $4(\frac{1}{4})^3 - 6(\frac{1}{4})^2 + \frac{13}{4} \cdot \frac{1}{4} - \frac{1}{2} = \frac{4}{64} - \frac{6}{16} + \frac{13}{16} - \frac{1}{2} = \frac{1}{16} - \frac{6}{16} + \frac{13}{16} - \frac{8}{16} = 0 \checkmark$

The other two roots are complex: $\beta = \frac{5}{8} \pm \frac{\sqrt{11}}{8}i$.

4.2 The parameter α^*

From eq. (4.16) of Castellani et al.:

$$\alpha_* = \frac{p^2}{q^2} (3\beta_* - 4\beta_*^2) \quad (12)$$

For $p = q = 1$, $\beta^* = \frac{1}{4}$:

$$\boxed{\alpha^* = 3 \cdot \frac{1}{4} - 4 \cdot \frac{1}{16} = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}} \quad (13)$$

4.3 The parameter γ^*

From Einstein equation (6):

$$\gamma^{*2} = \frac{3\nu/4}{\beta^*(1 - \beta^*)} = \frac{3\nu/4}{(1/4)(3/4)} = \frac{3\nu/4}{3/16} = 4\nu \quad (14)$$

Normalising $\nu = 1$: $\gamma^* = 2$.

4.4 Physical metric parameters

The physical squashing parameters a^2, b^2, c^2 are related to $\alpha^*, \beta^*, \gamma^*$ via the Killing metric normalisation on each factor of G_K . Castellani et al. use the normalisation:

$$N_{\mathbb{C}P^2} = \frac{24}{5}, \quad N_{S^2} = \frac{12}{5}, \quad N_{\text{fib}} = \frac{1}{20} \quad (15)$$

giving:

$$a^2 = N_{\mathbb{C}P^2} \cdot \alpha^{*2} = \frac{24}{5} \cdot \frac{1}{4} = \frac{6}{5} \quad (16)$$

$$b^2 = N_{S^2} \cdot (\gamma^* \beta^*)^2 = \frac{12}{5} \cdot (2 \cdot \frac{1}{4})^2 = \frac{12}{5} \cdot \frac{1}{4} = \frac{3}{5} \quad (17)$$

$$c^2 = N_{\text{fib}} \cdot (q\gamma^*)^2 = \frac{1}{20} \cdot (1 \cdot 2)^2 = \frac{1}{20} \cdot 4 = \frac{1}{5} \quad (18)$$

4.5 Verification

Summary of the Einstein metric on $M^{1,1,1}$:

$$\boxed{a^2 = \frac{6}{5}, \quad b^2 = \frac{3}{5}, \quad c^2 = \frac{1}{5}} \quad (19)$$

These satisfy the Einstein condition $R_{\alpha\beta} = 6 g_{\alpha\beta}$ (Sasaki–Einstein normalisation for $\dim = 7$).

Ratios: $a^2 : b^2 : c^2 = 6 : 3 : 1$, with $a^2/b^2 = 2$, $a^2/c^2 = 6$, $b^2/c^2 = 3$.

5. Identification with FisherGeometrics parameters

The metric parameters a^2, b^2, c^2 of the Einstein metric on $M^{1,1,1}$ are identified with the FisherGeometrics parameters as follows:

$$\tau = c^2 = \frac{1}{5} \quad (\text{radius squared of the } S^1 \text{ fibre}) \quad (20)$$

$$\kappa = a^2 = \frac{6}{5} \quad (\text{radius squared of the } \mathbb{C}P^2 \text{ factor}) \quad (21)$$

Theorem 2 (Derivation of τ and κ). *The FisherGeometrics parameters $\tau = \frac{1}{5}$ and $\kappa = \frac{6}{5}$ are uniquely determined by:*

1. *The requirement that the internal space K have isometry group $G_K = SU(3) \times SU(2) \times U(1)$ (Standard Model gauge group).*
2. *The requirement of $\mathcal{N} = 2$ supersymmetry, which selects $M^{1,1,1}$ uniquely among all $M^{p,q,r}$.*
3. *The unique real positive root $\beta^* = \frac{1}{4}$ of the cubic equation (4.15) of Castellani, D'Auria and Fré (1984), together with $\alpha^* = \frac{1}{2}$ from eq. (4.16).*

They are **not free parameters** of the FisherGeometrics framework but consequences of the Kaluza–Klein geometry.

6. Consequences for FisherGeometrics

6.1 Reduction of assumptions

The derivation reduces the independent inputs of FisherGeometrics from:

- **Before:** one postulate ($g_{AB} = \mathcal{F}_{AB}/\rho_0$) + two parameters (τ, κ) + one measurement (G_N)
- **After:** one postulate + one measurement (G_N)

The parameters τ and κ are now derived from the $\mathcal{N} = 2$ Einstein metric on $M^{1,1,1}$.

6.2 Connection to predictions

All predictions of FisherGeometrics that involve τ and κ now rest on the Kaluza–Klein geometry:

Observable	Formula	Derivation chain
$G_F = 1.1664 \times 10^{-5} \text{ GeV}^{-2}$	$\tau^2(n^2 + 1)/(4M_{\text{KK}}^2)$	$\tau = c^2(M^{1,1,1})$
$\eta = 6.14 \times 10^{-10}$	$(3/4)\tau^{13}$	$\tau = c^2(M^{1,1,1})$
$f_\pi = 92.2 \text{ MeV}$	$\Lambda_{\text{QCD}}\sqrt{N_c}/4$	via $\Lambda_{\text{QCD}}(\tau)$
$\lambda_W = 0.219$	$\tau\sqrt{\kappa}$	τ, κ from $M^{1,1,1}$

6.3 Open point

The Killing metric normalisation factors N_{CP^2} , N_{S^2} , N_{fib} connect the Castellani parametrisation $(\alpha^*, \beta^*, \gamma^*)$ to the physical metric (a^2, b^2, c^2) . These follow from the specific normalisation conventions of Castellani, D’Auria and Fré (1984), Nucl. Phys. B239, p. 619–623. An independent derivation of these factors from the FisherGeometrics Bures metric on $\mathcal{D}_6$ remains to be completed.

7. Primary sources

1. L. Castellani, R. D’Auria and P. Fré, *$SU(3) \times SU(2) \times U(1)$ from $D=11$ Supergravity*, Nucl. Phys. B**239** (1984) 610–652.
Equations used: (4.11), (4.13), (4.14), (4.15), (4.16), (4.17).
2. L. Castellani, L.J. Romans and N.P. Warner, *A Classification of Compactifying Solutions for $D=11$ Supergravity*, Nucl. Phys. B**241** (1984) 429–462.
Used for: uniqueness of $\mathcal{N} = 2$ solution (Table 1), definition of $M^{p,q,r}$ (Section 3).
3. D. Fabbri, P. Fré, L. Gualtieri and P. Termonia, *M -theory on $AdS_4 \times M^{1,1,1}$* , Nucl. Phys. B**560** (1999) 617–682. hep-th/9903036.
Used for: vielbeins (eq. 4.16), harmonic analysis (eq. 5.6–5.8).

FisherGeometrics Framework (June 2026). Working document. Speculative theoretical research.

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